



Operational Intelligence for Controlled Environment Agriculture

Varianz turns the raw telemetry of a controlled-environment greenhouse into clear, point-in-time operational intelligence. It continuously diagnoses **energy, resource and climate performance**, surfaces genuine operational deviations as they happen, and explains them in plain language — with every figure traceable to the underlying evidence. Varianz is built as **decision support**: it informs operators and investors, it does not actuate equipment or promise outcomes.

CAPABILITIES

What Varianz does

- **Energy & resource diagnosis** — daily heat, electricity (peak / off-peak), CO₂, irrigation and drain intensity, normalized against a weather-aware baseline.
- **Operational climate** — temperature, humidity, CO₂ and actuation tracked against realized setpoints and safe operating bands, with rolling compliance scoring.
- **Natural anomaly detection** — persistent, evidence-backed deviations (climate excursions, heat-baseline deviations, drain-ratio outliers), not synthetic threshold alerts.
- **Ask Varianz** — an evidence-grounded assistant that interprets the current state and recommends safe operator checks in plain language.
- **Point-in-time replay** — reconstruct any moment in the operating history for review, without ever leaking future data.
- **Ontario tariff economics** — an optional CAD cost view that stays hidden until a reviewed, sourced tariff profile is entered: no default prices, no savings claims.

STANDARDS & GOVERNANCE

Aligned with ISO 50001 & ISO 50006

Varianz is built around the methodology of the ISO 50000 energy-management family, giving energy managers an auditable data foundation:

- **ISO 50001 — Energy Management Systems.** Varianz supports the monitoring, measurement and analysis at the heart of an EnMS: continuous tracking of significant energy uses, versioned energy data, and an auditable trail for energy-performance evaluation and continual improvement.
- **ISO 50006 — Energy baselines & EnPIs.** Varianz establishes **weather-normalized energy baselines (EnB)** and **energy performance indicators (EnPI)** that adjust for the relevant variables (outdoor temperature, heating-degree hours, radiation, lighting hours, crop stage). Actual-vs-expected variance, uncertainty bands and robust deviation scoring follow the EnB / EnPI approach for like-for-like performance comparison.

Varianz provides the measurement methodology and evidence base to **support** these standards; it is a decision-support tool, not a certification.

HOW IT MEASURES PERFORMANCE

Machine-learning modeling

- **Weather-normalized baseline** — an ElasticNet regression models expected heat intensity from the relevant operational drivers, re-fit against a locked hold-out period for honest evaluation.
- **Disciplined model promotion** — the learned model is used only when it beats a naive rolling-median baseline by a required margin *and* passes cross-validation with no degraded fold; otherwise Varianz transparently falls back to the simpler baseline.
- **Robust anomaly scoring** — deviations are scored with a robust median/MAD z-score that resists outliers, and every estimate carries explicit p10–p90 uncertainty bands and a confidence level.



- **Honest scope** — forecasting and optimization are deliberately out of scope; nothing is presented as causal or as guaranteed savings.

THE VARIANZ ASSISTANT

Agentic capabilities

- **Evidence-grounded interpretation** — the assistant reasons only over a typed, curated evidence bundle drawn from the live snapshot, never over free-form data.
- **Verifiable claims** — every numerical statement must cite one or more evidence IDs; ungrounded answers are rejected and retried, then fail closed rather than guess.
- **Guardrails by design** — the agent separates observations, calculations, model estimates and recommendations; it never claims causality, promises savings, or implies equipment control, and treats all evidence as untrusted data.
- **Frontier models, structured output** — powered by current frontier LLMs (OpenAI GPT-5.6) with schema-validated, structured responses.

ENTERPRISE-READY FOUNDATIONS

Trust, governance & architecture

- **Full traceability** — every response carries data, definition and model versions plus the evidence IDs behind each figure.
- **Multi-tenant by design** — a Supabase / PostgreSQL system of record with row-level security isolating each organization's data.
- **Decision-support only** — Varianz informs; it does not actuate. No physical control, by design.

See your operation the way your data does.

Varianz runs against a full controlled-environment reference deployment — the Wageningen greenhouse dataset, over one million validated observations — so every capability above is demonstrated on real operating data. Get in touch to arrange a guided walkthrough.

[Request a guided walkthrough of the Varianz reference deployment.](#)